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Department of Computer Science and Business Systems
Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: B.Tech., IV & CSBS
Course Code & Title: AL3451 & Machine Learning
Name of the Faculty member (s): Ms.M. Preethi Ram, AP/CSBS

Innovative Practice Description

- **Unit / Topic: Unit 4/ Batch Normalization**
- **Course Outcome:** CO4
- **Activity Chosen:** Think –Pair Share

- **Justification:**

  Batch Normalization is a technique used in machine learning to improve the training performance and stability of neural networks by normalizing the inputs to each layer. The Think-Pair-Share activity helps students grasp this concept by first prompting them to think individually about why normalization is necessary and how it affects learning rates and internal covariate shift. They then pair up to discuss their ideas, clarify doubts, and compare interpretations. Finally, sharing with the whole class promotes a deeper understanding of where Batch Normalization fits within the training process, its impact on convergence speed, and its effect on model generalization. This collaborative learning method encourages critical analysis, peer teaching, and consolidation of complex theoretical and practical aspects of the technique.

- **Time Allotted for the Activity:** 35 Minutes

- **Details of the Implementation:**

A Collaborative, active learning strategy, in which students work on a question posed by

1. Think Phase (5 minutes):
Students were first asked to individually reflect on the concept of Batch Normalization. They wrote brief responses to prompts such as:

- What problem does Batch Normalization solve in deep learning?
- How does normalizing activations during training help with convergence?
- What are the key components (mean, variance, scale, shift) involved in Batch Normalization?

This phase allowed students to activate prior knowledge and identify areas where they needed clarity.

2. Pair Phase (10 minutes):
Students then paired up and discussed their answers with a partner. During this phase, they:

- Compared their individual responses.

- Clarified each other's doubts and misconceptions.
- Used diagrams and equations from their notes to explain concepts. This collaborative step helped reinforce theoretical understanding through peer explanation.

3. Share Phase (10–15 minutes):
After the pair discussions, selected pairs shared their insights with the entire class. The instructor facilitated the discussion by:

- Asking probing follow-up questions like "How does Batch Normalization affect the learning rate or backpropagation?"
- Highlighting accurate interpretations and addressing common errors.
- Summarizing key points on the board and linking the discussion to real-world examples and training graphs.

- **CO – PO / PSO mapping:**

CO	**PO1**	**PO2**	**PO3**	**PO4**	**PSO1**	**PSO2**
CO 4	2	2	2	1	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PSO1**	**PSO2**
	2	2	2	1	2	2
Justification for correlation	students apply mathematical and statistical principles (like mean and variance normalization) to real machine learning models.	students analyze issues like vanishing/exploding gradients or slow convergence in deep networks and consider Batch Normalization as a solution.	Students integrate Batch Normalization into model architectures to meet specific performance or training time requirements.	students experiment with and analyze the impact of Batch Normalization on model performance using datasets, tuning hyperparameters and comparing results.	**PSO1** is directly mapped since Batch Normalization is a key technique in **Machine Learning/Deep Learning**, aiding in the deployment of real-world AI solutions in finance, healthcare, manufacturing, and more.	**PSO2** is relevant as students learn to apply **computational statistics**—specifically normalization an variance reduction—to improve model performance and robustness.

- **Images / Screenshot of the practice:**

 *Fig.1. Discussion done by students for the given topic.*	 *Fig.2. Kathirvel shared his ideas to her classmate.*

Reflective Critiques

❖ **Feedback of practice from students and other stakeholders:**

❖ Some of the students write answers for the questions easily and share their ideas with others. Some of them need more explanation and examples to get clarity.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to share ideas with classmates and builds oral communication skills. They completed the task by coordinating with each other and get more confidence, clear understanding in this topic when discussing with their peers.

❖ **Challenges faced in implementation:**

- The students who didn't understand the concept clearly is not able to complete it within time.
- Making all the students to involve and present actively in this activity is a challenging.

References:

❖ https://www.ritrjpm.ac.in/images/computer-science/CS8451-Think-Pair-Share_PJT.pdf

❖ https://www.youtube.com/watch?v=1yRJf8j8xVU